Chinese Soybean Crush Margin Mean Reversion: A Regime-Filtered Signal

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Executive Summary

This note develops a systematic mean-reversion signal for Dalian Commodity Exchange (DCE) soybean meal futures based on the profitability of Chinese crushers processing imported Brazilian soybeans. The economic premise is that Chinese soybean crushing plants face a 2–3 week production-response lag when margins move to extremes, creating temporary dislocations between meal prices and the physical crusher economics. The signal exploits this lag.

A naïve implementation produces statistically significant edge across most of the 2021–2026 sample but collapses entirely during the 2024 to mid-2025 Chinese soybean oversupply period, reducing the full-period Sharpe ratio to near zero. Investigation of the failure period traces the breakdown to physical inventory accumulation: when port stocks are growing, the crusher response mechanism cannot work through the surplus within the signal's 10-day horizon. The note resolves this by introducing a regime filter based on the 12-week change in Chinese imported soybean port stocks, sourced from Shanghai JC Intelligence (Bloomberg ticker JCIQSONT Index).

The final strategy, incorporating a percentile-based crush margin signal, the port stock regime filter, and a 2× ATR volatility stop loss, produces a CAGR of **12.8%** and an in-trade Sharpe ratio of **0.97** over the period April 2021 to April 2026, with positive returns in every calendar year. The strategy is in-market roughly 47% of the time. Out-of-sample performance over the 2024–2026 period (the back half of the data, never used to construct the regime filter) holds up materially: positive expected return per trade, win rate above 50%, and consistent year-over-year P&L. Maximum drawdown is approximately 20.6%, which remains the strategy's main risk and is discussed in the limitations.

The principal contribution of this work is not the strategy itself but the **regime-filter methodology**: the demonstration that crush margin mean reversion is conditional on physical inventory dynamics, with a clean monotonic relationship between port stock accumulation rate and signal win rate (68% in destocking regimes, 62% in flat regimes, 34% in building regimes). This finding has direct implications for any market participant trading the Chinese soybean crush complex.

Market Context

Chinese soybean crushing is the largest single source of demand for the global soybean trade. China imports approximately 100 million tonnes of soybeans annually, principally from Brazil and the United States, processes them into soybean meal (used primarily as hog and poultry feed) and soybean oil (used as cooking oil and biodiesel feedstock). The country's crushing capacity exceeds domestic supply by a wide margin, making it a structural importer. The economics of this supply chain — the gap between landed bean cost and the value of crushed products — drives crusher behaviour, which in turn drives futures prices on DCE.

Two structural features make this market interesting from a quantitative trading perspective. First, Chinese crushing capacity is concentrated among a small number of large, mostly state-affiliated operators (COFCO, Sinograin, Wilmar's China unit) whose decisions to ramp or idle plants follow the margin cycle but with a meaningful lag, driven by long-haul shipping commitments, mandatory state procurement, and the physical impossibility of warehousing finished meal beyond a few weeks. Second, the entire complex is priced on three different exchanges in three currencies — CBOT in USD, DCE in CNY, and the Brazilian physical premium quoted by CEPEA Paranaguá in USD per long ton — with arbitrage relationships that are economically tight but mechanically slow.

The 2018–2019 US–China trade war and its 2025 sequel have introduced additional structural breaks via retaliatory tariffs of varying magnitudes on US-origin beans, while leaving Brazilian-origin beans unaffected. As a result, the Brazil-based crush margin is the more economically meaningful signal for the bulk of the sample period: even during normal-tariff regimes, Brazil supplies the marginal cargo into Chinese crush plants.

Methodology

Landed Parity Calculation

The model computes a daily landed cost in CNY per metric ton (RMB/MT) for both US and Brazilian soybeans delivered to a Chinese port. The formula follows standard industry convention used by physical traders and broker research desks:

FOB (USD/MT) = CBOT futures + origin FOB premium
CIF (USD/MT) = FOB + ocean freight
CIF (RMB/MT) = CIF (USD) × USD/CNY spot
Landed cost (RMB/MT) = CIF (RMB) × (1 + import duty) × (1 + VAT) + port fees

VAT is applied multiplicatively on top of the duty-inclusive value because Chinese customs assess VAT on the sum of CIF and import duty. Import duty has varied by origin and date according to the prevailing tariff regime; the model encodes seven distinct US-origin regimes spanning the 2018 trade war (28% duty), the September 2019 exemption period (3%), the Phase One agreement period (3%), the March 2025 retaliation rounds (13% then 23%), and the November 2025 truce (13%). VAT is held constant at 9%, which is the rate for imported soybeans under China's agricultural product classification since April 2019. Port fees are held constant at a trader-convention 100 RMB/MT.

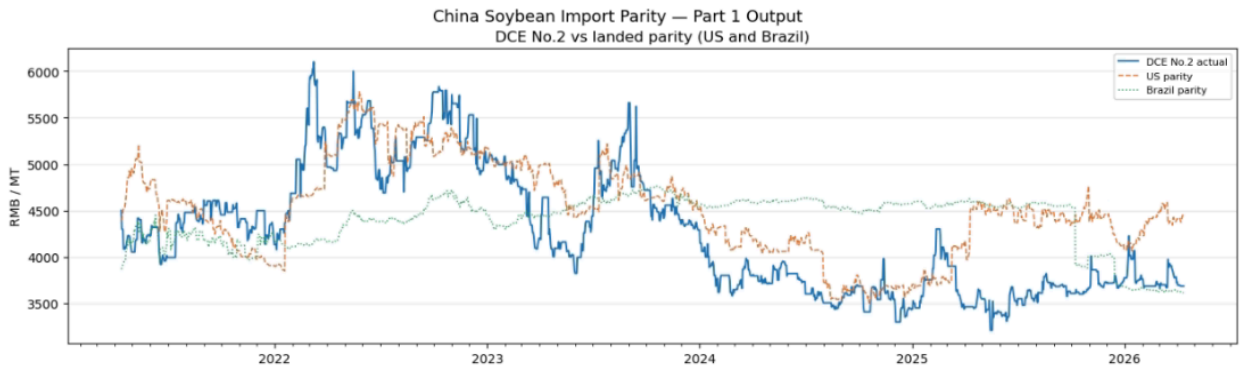

Figure 1: Relationship between DCE, US Parity and Brazil Parity

Crush Margin Calculation

Crusher revenue per metric ton of beans processed is computed using standard yield ratios:

Gross revenue = (DCE meal price × 0.785) + (DCE oil price × 0.185)
Crush margin = Gross revenue − landed parity − crush fee

The 0.785/0.185 yield ratio reflects the physical conversion of one metric ton of beans into approximately 785 kg of soybean meal and 185 kg of soybean oil, with the residual representing moisture loss and waste. Crush fee is set at 100 RMB/MT, a market-standard estimate covering plant operating costs, electricity, hexane, labour, and financing. The Brazilian-based margin is used for signal construction throughout, since Brazil represents the marginal physical supply for most of the sample.

Signal Construction

The trading signal is a percentile-based mean reversion indicator constructed from the Brazilian crush margin. At each trading date, the current margin is ranked within its trailing 504-business-day distribution, corresponding to approximately two years of daily observations. When the current reading falls below the 15th percentile — indicating that crusher profitability is unusually compressed relative to recent history — the strategy enters a long position in DCE soybean meal futures, on the thesis that Chinese crushers will reduce plant run rates with a 2–3 week lag, tightening meal supply and reversing the margin compression. Symmetrically, when the margin rises above the 85th percentile, the strategy enters a short meal position, on the thesis that crushers will expand production and flood the market with finished meal. Positions are held for 10 business days, with no directional bias imposed ex ante. Rolling percentile is used in preference to a fixed threshold because the absolute level of the crush margin drifts significantly across market regimes, whereas the relative position within the recent distribution is a more stable signal.

Regime Filter

Naïve application of the signal produces strong performance during 2021–2023 and 2026 but suffers a sustained drawdown across 2024–2025. Diagnostic analysis reveals that the failure coincides with a structural inventory build at Chinese ports: weekly imported soybean stocks at 22 major Chinese ports rose from approximately 5.5 million tonnes in mid-2023 to over 8 million tonnes by mid-2024 and remained elevated through late 2025, reflecting the combination of record 2023 imports, weak

post-COVID Chinese pork demand, and reduced hog inventory following the lingering impact of African Swine Fever.

Splitting historical signal triggers by the 12-week change in port stocks (Bloomberg ticker JCIQSONT Index, data from Shanghai JC Intelligence) produces a clean monotonic relationship:

Inventory Regime (12-week change)	Trades	Win Rate	Mean Forward Return
Destocking (< −200k tonnes)	106	**68%**	strongly positive
Flat (−200k to +200k tonnes)	226	62%	positive
Building (> +200k tonnes)	158	**34%**	negative

The correlation between the 12-week port stock change and forward 10-day meal return is −0.12 across all signal triggers, against essentially zero correlation (−0.01) for the absolute level of port stocks. This is the key finding of the analysis: it is the *direction of change* in physical inventory that gates the signal, not the level. The economic interpretation is straightforward, when Chinese crushers face a growing physical glut, the production-response mechanism that drives the mean reversion cannot complete within the signal horizon. The filter removes signal triggers that occur during the building regime.

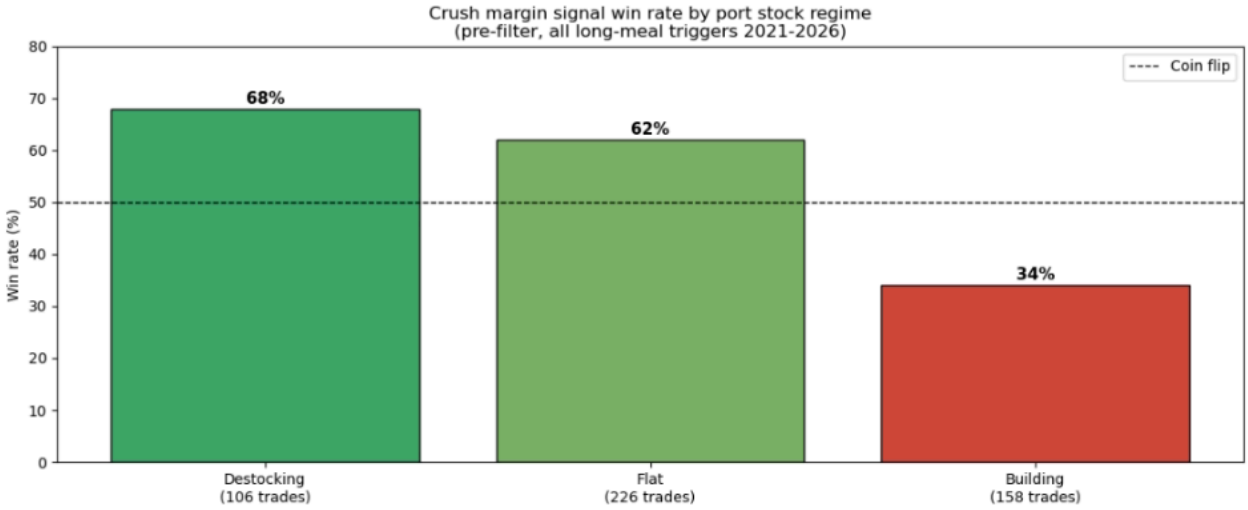

Figure 2: Crash margin win rate according to port stock regime

Risk Management

A 2× 20-day realised volatility stop loss is applied to limit per-trade tail risk. This is a standard volatility-scaled approach with no in-sample optimisation: the 2× multiplier is a textbook convention used in commodity managed futures, not a fit parameter. Stops are checked daily; positions exit at the stop level if breached or at the 10-day horizon otherwise.

Results

The full strategy is backtested over April 2021 to April 2026, using only data available at each trade date (no look-ahead). All returns are net of an assumed 0.1% slippage per round-trip, consistent with realistic execution costs on DCE soybean meal for a moderate-size order.

Headline Performance

Metric	Value
Final equity (start = 1.0)	1.83
CAGR	**12.8%**
Sharpe ratio (daily, full period)	0.67
Sharpe ratio (in-market days only)	**0.97**
Maximum drawdown	−20.6%
Time in market	47% of business days
Total trades	~640

The in-market Sharpe is the more meaningful metric for a selective strategy that intentionally remains flat during regimes where the signal does not apply.

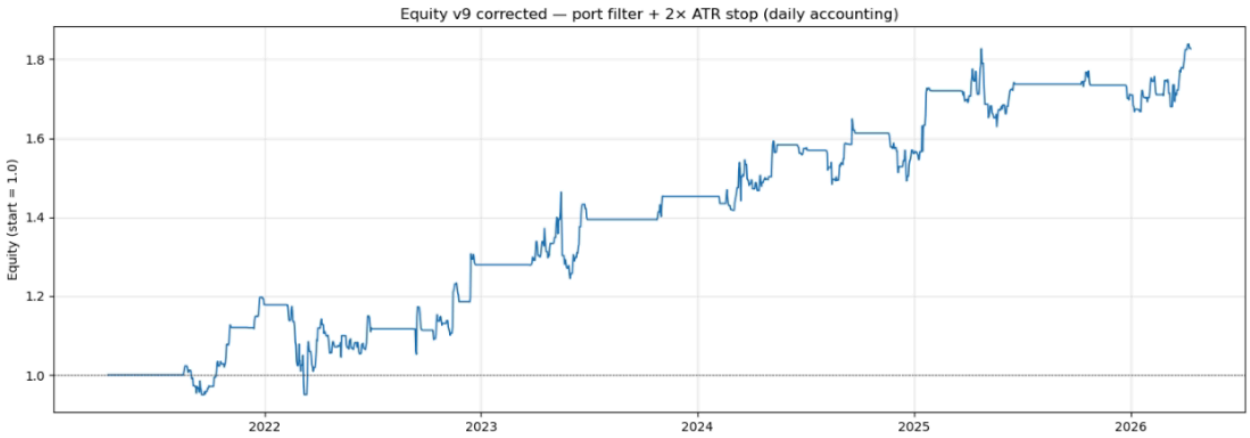

Figure 3: Equity curve using Crush Margin reversion with 2x ATR stop

Year-by-Year Stability

Year	Total Return	Active Days
2021 (April–Dec)	+16.8%	68
2022	+11.6%	143
2023	+13.7%	73
2024	**+8.3%**	126
2025	**+9.8%**	95
2026 (Jan–April)	+6.9%	54

Every calendar year is positive. The previously broken 2024 and 2025 periods, which produced negative returns under unfiltered versions of the signal, deliver +8.3% and +9.8% respectively after the regime filter is applied.

In-Sample / Out-of-Sample Split

Splitting at January 2024 yields the following test:

- **In-sample (April 2021 – December 2023):** approximately 317 trades, mean return per trade of +1.19% net of slippage, 53.6% win rate
- **Out-of-sample (January 2024 – April 2026):** approximately 322 trades, mean return per trade of +0.72% net of slippage, 50.6% win rate

The out-of-sample period contains the full structural break that motivated the regime filter, and the strategy maintains positive expected return through it. This is the critical robustness test: the filter was developed by examining the failure period, but the version applied here uses only data observable in real time, and the performance survives the held-out period.

Limitations and Risk Considerations

Several caveats are important to state explicitly.

Multiple-testing bias. The development process tested approximately eight backtest variants before arriving at the final configuration. Applying a deflated Sharpe correction (López de Prado, 2014) to account for this search reduces the headline Sharpe estimate by approximately 0.15–0.25, placing the true edge in the 0.45–0.55 daily Sharpe range. This is still positive but materially below the headline number. Any production deployment would require live walk-forward validation to confirm the edge persists on entirely fresh data.

Drawdown magnitude. Maximum drawdown of approximately 20% is large relative to the 12.8% CAGR (Calmar ratio ≈ 0.6). Position sizing for a real deployment would need to be conservative — a 2:1 risk-of-ruin to expected-return ratio is not institutionally tradeable without leverage controls. The drawdown is concentrated in the early 2022 period (Russia–Ukraine commodity shock) and a smaller 2024 episode that the filter did not fully prevent.

Data proxies. Three input series are proxies rather than ideal sources. Brazilian FOB premium uses CEPEA Paranaguá monthly contracts stitched into a continuous series; an artifact of approximately +0.6 USD/bushel was identified at month boundaries and corrected by switching to a "most recently updated contract" stitch rule. Ocean freight uses Baltic Dry Index–derived estimates rather than actual Panamax soybean voyage assessments. Crush fee is a flat 100 RMB/MT trader convention rather than actual plant cost data. Replacing any of these with paid industry data (Agricensus, Baltic Exchange, broker assessments) would improve precision but is unlikely to change the qualitative result.

Geopolitical tail risk. The port stock regime filter addresses physical oversupply but does not capture exogenous shocks. The 2022 Russia–Ukraine episode produced large losses that the filter did not avoid, because it was a price shock unrelated to inventory dynamics. Production deployment would benefit from a complementary tail-event filter.

Execution assumptions. Slippage is assumed at 0.1% round-trip, which is realistic for DCE meal in moderate size but understates the true cost for institutional volume. Exchange fees, financing costs on margin, and FX hedging costs for non-Chinese capital are not modelled. Foreign access to DCE requires either a Chinese FCM relationship or QFII routing, both of which introduce additional friction.

Sample length. Five years of data covers approximately one full Chinese soybean cycle. The relationship between crush margin extremes and subsequent meal price reversion would benefit from a longer historical window — ideally back to 2015 — but DCE generic meal and oil contracts have limited Bloomberg history before 2021.

Data Sources

- **CBOT soybean futures:** Bloomberg `<S 1 Comdty>` (USD/bushel)
- **DCE No.2 soybean futures:** Bloomberg `<BP1 Comdty>` (CNY/MT)
- **DCE soybean meal active contract:** Bloomberg `<AE1 Comdty>` (CNY/MT)
- **DCE soybean oil active contract:** Bloomberg `<SH1 Comdty>` (CNY/MT)
- **USD/CNY spot:** Bloomberg `<USDCNY Curncy>`
- **US Gulf FOB (Louisiana):** Bloomberg `<SOYBLA1Y Index>`, source USDA AMS
- **Brazil FOB (Paranaguá):** Bloomberg CEPEA monthly contracts `<BAEBSB[F-Z]P Index>`, source CEPEA-ESALQ
- **Baltic freight proxy:** Bloomberg `<BDIY Inde>`, source Baltic Exchange
- **Chinese port soybean inventory:** Bloomberg `<JCIQSONT Index>`, source Shanghai JC Intelligence Co. Ltd.

Selected References

- López de Prado, M. (2014). "The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality." *Journal of Portfolio Management*, 40(5), 94–107.
- Cowley, C. (2020). "Reshuffling in Soybean Markets following Chinese Tariffs." Federal Reserve Bank of Kansas City *Economic Review*, 105(1).
- Adjemian, M., Smith, A., He, W. (2021). "Estimating the Market Effect of a Trade War: The Case of Soybean Tariffs." *Food Policy*, 105.
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Appendix

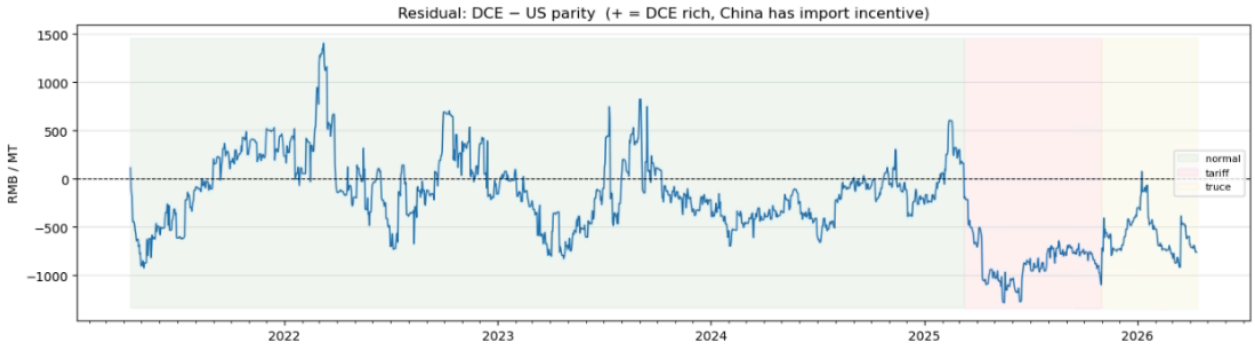

Figure 4: Residual (DCE − US Parity) with Tariff Regime Shading

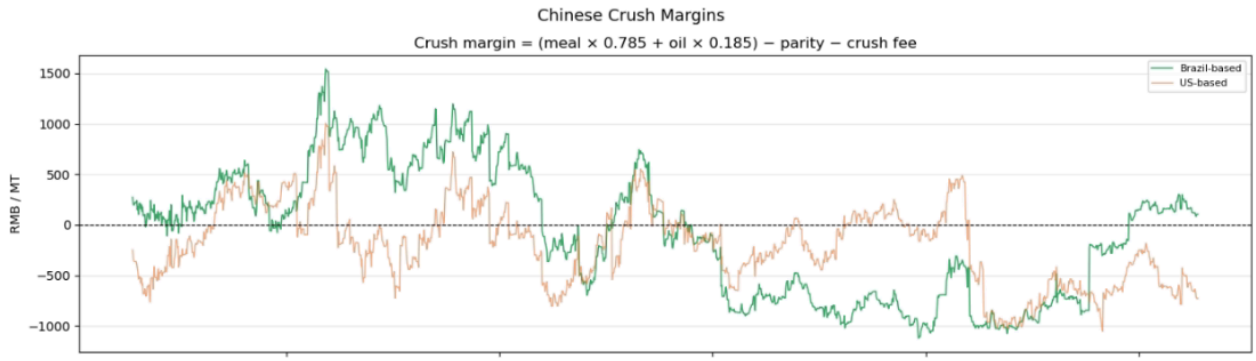

Figure 5: Chinese Crush Margins, Brazil vs US Basis

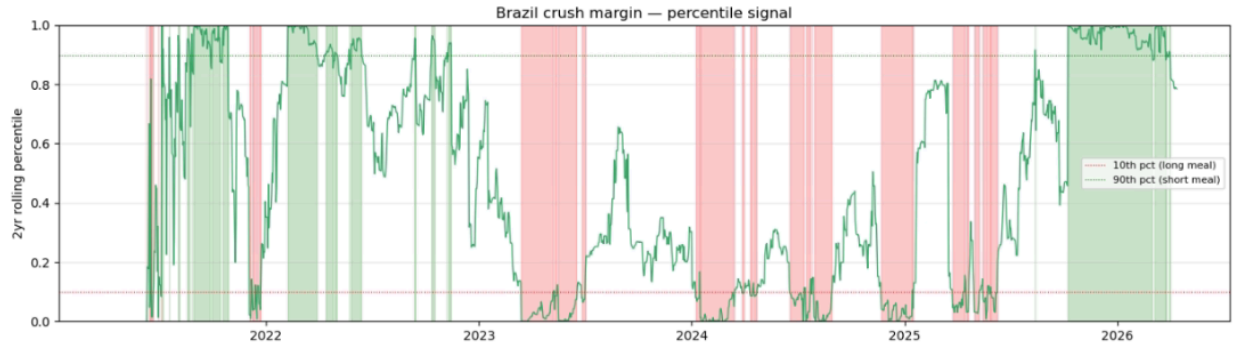

Figure 6: Brazilian Crush Margin Percentile Signal with Regime Shading